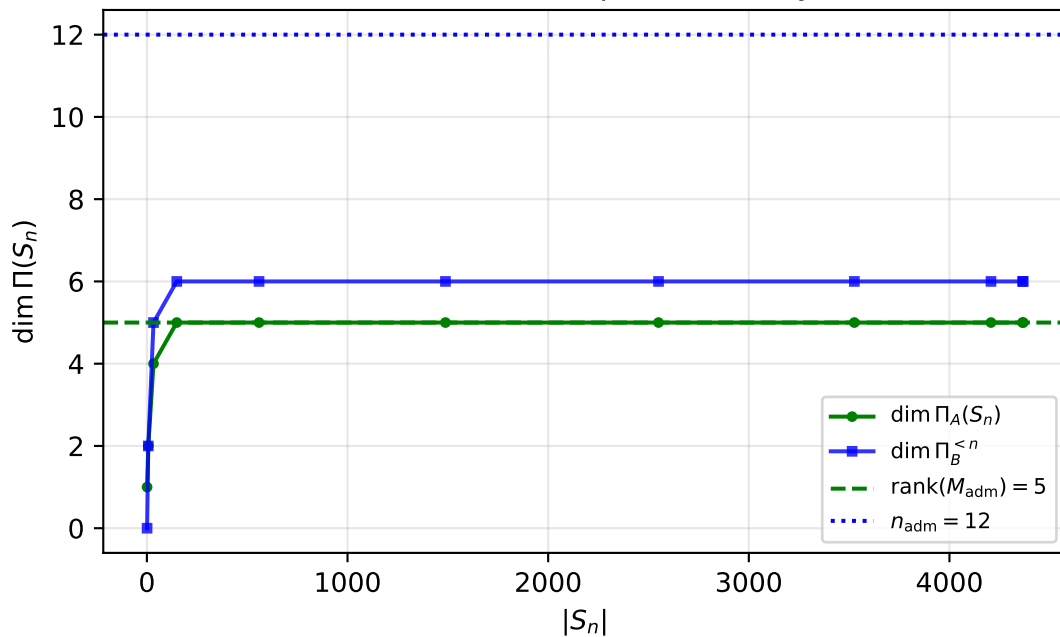
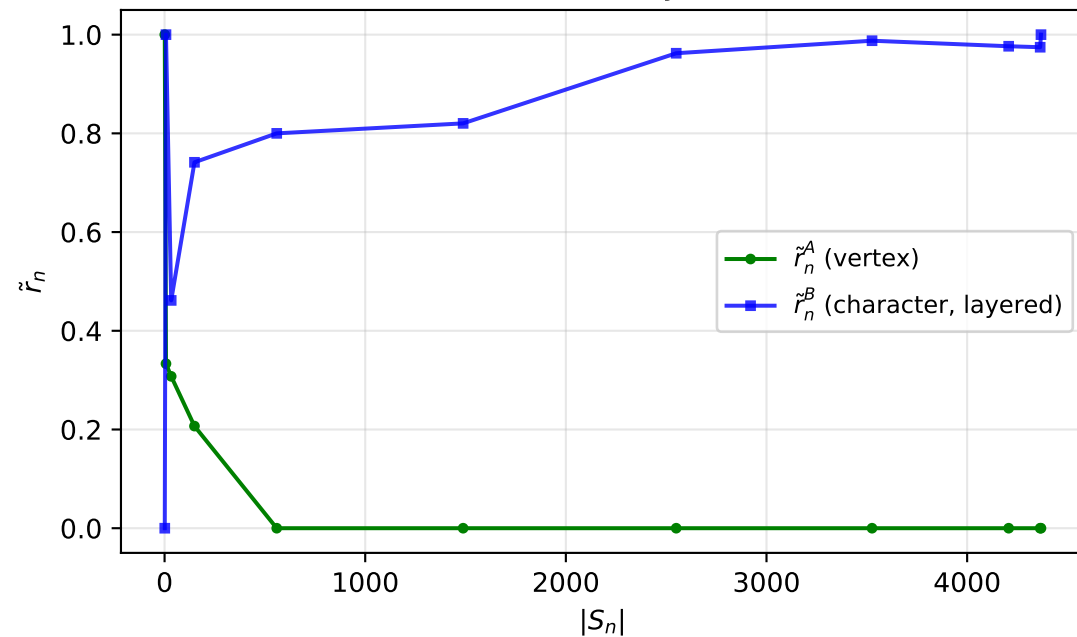


Version A vs B (character-based) on  $X^{5,13}$ , shell-layered cascade  
 $|G| = 4368$ ,  $n_{\text{adm}} = 12$ ,  $\text{rank}(M_{\text{adm}}) = 5$

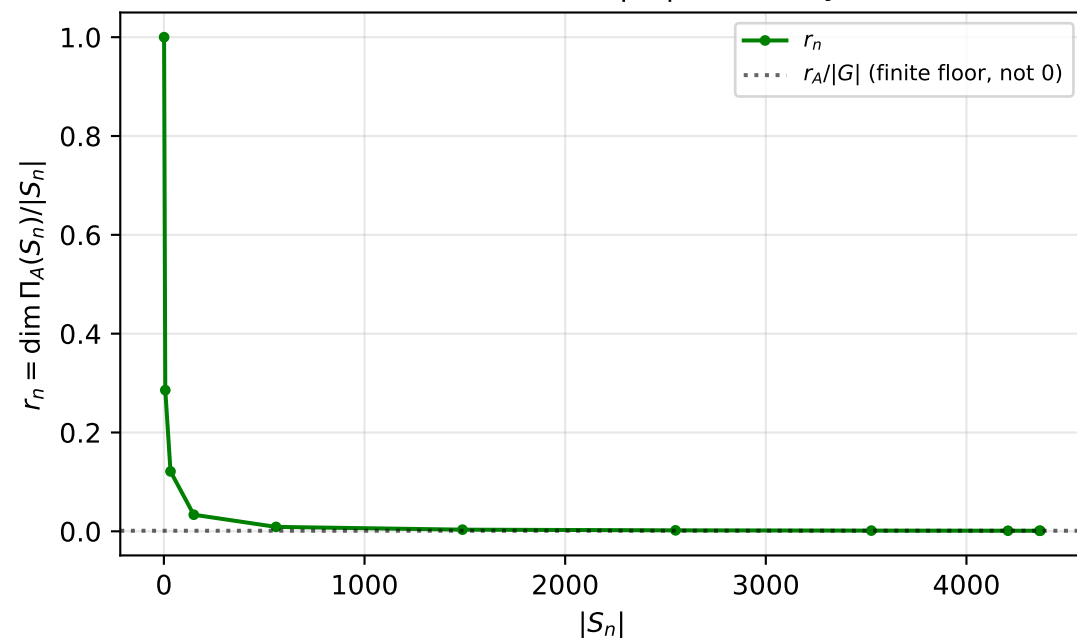
Rank of admissible span (shell-layered)



Shell-wise novelty fraction



Finite bound  $r_n \leq r_A/|S_n|$  (Corollary~1)



Summary

KEY NUMBERS ( $q = 13$ ,  $p = 5$ ), shell-layered:

$|G| = 4368$

$n_{\text{adm}} = 12$

$\text{rank}(M_{\text{adm}}) = r_A = 5$

VERSION A (empirical, this traversal only):

dim  $\Pi_A(S_n)$  reaches  $r_A$  at  $|S_n| \approx 149$

(abstract witness only guarantees size  $r_A = 5$ ;

the gap is the BFS/shell discovery cost, not a bound)

Late  $\tilde{r}^A$ : 0.000

VERSION B (character, layered):

Max dim  $\Pi_B^{\leq n}$ : 6

Late  $\tilde{r}^B$ : 0.980

(no decay -- character collapse persists  
under the layered definition too)